

Predicting IDH Mutation in Gliomas Using Attention-based Multiple Instance Learning

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Abstract. A glioma is a brain tumor in the central nervous system. Although it is a relatively rare type of cancer, it is also one of the most fatal ones. One biomarker for gliomas is isocitrate dehydrogenase (IDH). In particular, gliomas can be either IDH-mutated or IDH-wildtype, and the cancer’s prognosis, aggressiveness, and appropriate treatment plans largely depend on the IDH mutation status. Now, identifying the IDH mutation status normally requires invasive approaches such as surgery. As such, there is a need to explore potential non-invasive approaches. Here, we applied deep learning on magnetic resonance imaging (MRI) scans to predict the IDH mutation status among glioma patients. Since a single MRI scan can consist of multiple images or slices, we leveraged attention-based multiple instance learning to simultaneously consider all slices in prediction. Moreover, we explored incorporating radiomics-based features, as well as patient demographics, in the neural network classifier. Multiple MRI contrasts and clinical data on 625 patients were obtained from the USTW-Glioma dataset taken from The Cancer Imaging Archive. Ultimately, we compared the performance across different configurations, such as using different contrasts, different combinations of features, and so on. Based on the ablation study and comparisons, the classifier attains the best performance when all contrasts are utilized and combined with both radiomics-based features and patient demographics, achieving an ROC-AUC of about 0.90, a macro PR-AUC of about 0.89, and an overall accuracy of about 87%.

Keywords: multiple instance learning · radiomics · IDH mutation

1 Introduction

As mentioned in [16], a glioma is a brain tumor in the central nervous system, particularly originating from the glial cells, and is the most common type of brain tumor. Gliomas are relatively rare, affecting just about 0.006% of the population in the United States [16], but they form 81% of all malignant brain tumors and are often particularly fatal [18].

One biomarker for gliomas is isocitrate dehydrogenase (IDH). In general, gliomas can be either IDH-mutated or IDH-wildtype (i.e., they can be characterized by their *IDH mutation status*), with appropriate diagnostic and treatment strategies varying based on the mutation status [4]. Usually, IDH-mutated gliomas show a more favorable prognosis compared to IDH-wildtype ones [9]. In connection with this, identifying the IDH mutation status is crucial for preparing treatment plans [17]. Unfortunately, identification of the IDH mutation status requires invasive approaches, such as surgery [4].

As such, coming up with non-invasive approaches may be beneficial. This is where deep learning can help. By applying machine learning on magnetic resonance imaging (MRI) scans directly or on radiomics-based features [4,22,26,28], the IDH mutation status can possibly be predicted without requiring surgeries and biopsies. In this paper, we introduce a computer vision model for IDH mutation status classification. Particularly, we leverage attention-based multiple instance learning directly on MRI scans and consider different combinations of contrasts, radiomics-based features, and demographics, forming an ablation study and comparisons.

The rest of the paper is organized as follows. In Section 2, we briefly go over literature related to the topic at hand. Next, in Section 3, we describe our methodology, including data preparation, model architecture, and more. Finally, in Section 4 and Section 5, we discuss the findings, particularly those related to model performance, and summarize with key results and recommendations.

2 Related Works

Leveraging deep learning on medical images for molecular classification (e.g., predicting IDH mutation status) is not new. For example, [2] provided a systematic review of applications of machine learning or deep learning on histopathological images for classifying molecular markers in gliomas, identifying more than 2400 reports and ultimately examining 22 relevant studies. They found that deep learning models show great promise for predicting IDH mutation status, obtaining a pooled average of 0.8546 for accuracy and of 0.8653 for ROC-AUC (i.e., the area under the Receiver Operating Characteristic curve). Although they noted that such deep learning models should be considered only as “complementary tools” instead of stand-alone diagnostic devices, given the current status of research in the field (e.g., limited datasets, no standardized reporting).

Several studies focused on applying computer vision methods directly on medical images (e.g., histopathological images and MRI scans). For instance, [7] developed a convolutional neural network with multiple-instance learning on histopathological images of glioma tissues for predicting IDH mutation status, achieving an ROC-AUC of 0.84. Later works such as [13] applied newer methods, such as the vision transformer, on the same type of medical images for predicting IDH mutation status and achieved an even better ROC-AUC value of 0.96.

Aside from histopathological images, MRI scans have also been considered for IDH mutation status prediction with deep learning. For example, [4] lever-

aged nnU-Net to predict IDH mutation status based on MRI scans, obtaining probability estimates for classification on each slice and performing majority voting to ultimately predict at a patient level, achieving an ROC-AUC of 0.94 to 0.96. The authors noted that previous approaches [8,26,28] performed classification at a slice level (as opposed to patient level, as in their approach), which suffered from data leakage (since slices from the same subject are presumably correlated).

Still on MRI scans, aside from convolutions or transformers, radiomics-based features have also been leveraged for classification with deep learning. In general, radiomics is a field concerned with extracting quantitative features from medical images, such as tissue shape, lesion characteristics [15]. For example, [22] considered various radiomics-based models to predict IDH mutation status using MRI scans and achieved an ROC-AUC of 0.68, and even developed a radiomics risk score for prognosis prediction. Another study from [10] leveraged radiomics-based features from MRI and used traditional classifiers (e.g., support vector machines and random forests) to achieve an ROC-AUC of about 0.83. Finally, radiomics-based features have also been explored for predicting IDH mutation status in other medical image types such as computed topography (CT) scans (e.g., 0.85 ROC-AUC from [17]).

Now, in medical imaging, there are typically numerous “slices” or “images” that result from one scan. For example, in MRI and CT scans, there are multiple “slices” across an axis that, when combined, can give a picture of the 3D nature of the brain. Unsurprisingly, multiple instance learning (MIL) has been a popular method for deep learning with medical images. For example, [2] reviewed several studies employing attention-based MIL, such as [7] and [12]. More recently, [25] developed a hierarchical attention-based multiple instance learning method for predicting IDH mutation status using MRI scans, achieving an ROC-AUC of 0.89 to 0.92.

In this work, we also leverage attention-based multiple instance learning, particularly from [11], on MRI scans (as opposed to histopathological images, which usually require invasive approaches to obtain). Similar to [4], we operate at a patient level, representing each patient as a “bag” of MRI scans. However, we primarily focus on leveraging transfer learning by using a pre-trained backbone model as a feature extractor and combining such with radiomics-based and demographics-based features, and performing an ablation study on different configurations to unify different feature groups.

3 Methodology

3.1 Dataset

The dataset used was the *UTSW-Glioma* dataset from [21]. It is available for download and public access at The Cancer Imaging Archive [23]. The dataset contains multi-contrast MRI images (i.e., pre-contrast-T1w, post-contrast-T1w, T2w, and T2w-FLAIR) from 625 glioma patients, with additional information on

demographics (e.g., age, sex, and race) and clinical characteristics (e.g., IDH mutation status and tumor grade). Segmentation masks, both automated ones from the Federated Tumor Segmentation (FeTS) [20] and those manually corrected by experts, were also available for most patients. For extracting radiomics-based features, the FeTS-generated segmentation masks were utilized.

Overall, 622 patients had data on their IDH mutation status and of those, 619 had segmentation masks. Because IDH mutation status is the target variable, only those 622 patients were considered in the study. Moreover, since segmentation masks were necessary for extracting radiomics-based features, only those 619 patients were considered for all models that required radiomics-based features. This means that the “whole” dataset for configurations that excluded radiomics consisted of the 622 patients while those of configurations that included radiomics consisted of the subset of 619 patients.

Out of the 622 patients, 176 had IDH mutations while the rest did not (i.e., IDH-wildtype). The $622 - 619 = 3$ patients that did not have segmentation masks all belonged to the IDH-wildtype class. Here, we considered IDH-mutated to be the positive class. The datasets (622 for without radiomics and 619 for with radiomics) were split into three smaller subsets in a stratified manner (based on the target variable): 70% for training, 15% for validation, and 15% for testing.

Each MRI scan for a patient consisted of 155 slices along the axial plane. Note that splitting was done at a patient level, not at a slice level, where a single patient was represented by a “bag” of 155 images (as in multiple instance learning). The USTW-Glioma dataset provided versions of the MRI scans that were pre-processed using the Advanced Normalization Tools (ANTs) software from [3]. Such versions were utilized in the study to ensure standardization across slices, contrasts, and patients.

3.2 Feature Extraction

For extracting features from MRI slices (i.e., MR images), the BiomedCLIP, a vision transformer foundational model, from [27] was used. That is, each MR image was embedded as a 512-dimension vector using BiomedCLIP. Whenever multiple contrasts are used, each set of corresponding MR images (e.g., the i th slice from each contrast) was first embedded as a 512-dimension vectors separately. Then, the individual embeddings were concatenated to form the embedding for the multi-contrast slice or instance. For example, if all 4 contrasts were used, then each slice was embedded as a 2048-dimension vector, which meant that each bag was embedded as a 155×2048 feature map (where 155 is the number of instances or slices). If only one contrast was used, then each bag was embedded as a 155×512 feature map.

Meanwhile, radiomics-based features were extracted from the T2w-FLAIR contrast at a patient level (i.e., bag level). Only the following radiomics features, as defined in [30], were used:

1. Flatness (N17B)
2. Least axis length (7J51)

3. Short runs emphasis (22OV)
4. Dependence count variance (DNX2)
5. Normalized dependence count non-uniformity (OKJI)

Based on [17], these five were possibly the most useful radiomics-based features for IDH mutation status prediction.

Finally, for demographics features, only age at imaging (in years) and sex at birth (i.e., female or male) were considered.

3.3 Model Architecture

The main model architecture (i.e., a neural network) used here is shown in Figure 1. After extracting the feature map using the (frozen) backbone (i.e., Biomed-CLIP), the embeddings were projected into a smaller number of dimension with a linear layer. Then, we had a multi-head attention block using the gated attention mechanism described in [11]. Next, using the attention weights, the feature map was pooled and flattened into a 1 by KP feature vector, where K is the number of attention branches and P is the projection size in the first linear layer. For configurations with radiomics-based features and / or demographics features, the feature vectors were simply concatenated. Ultimately, the final feature vector was fed into a classifier head composed of a linear layer with a rectified linear unit (ReLU) activation function (for non-linearity) and a final linear layer to produce a logit estimate for the positive class. The neural network was trained with the binary cross-entropy loss function using the AdamW optimizer from [14].

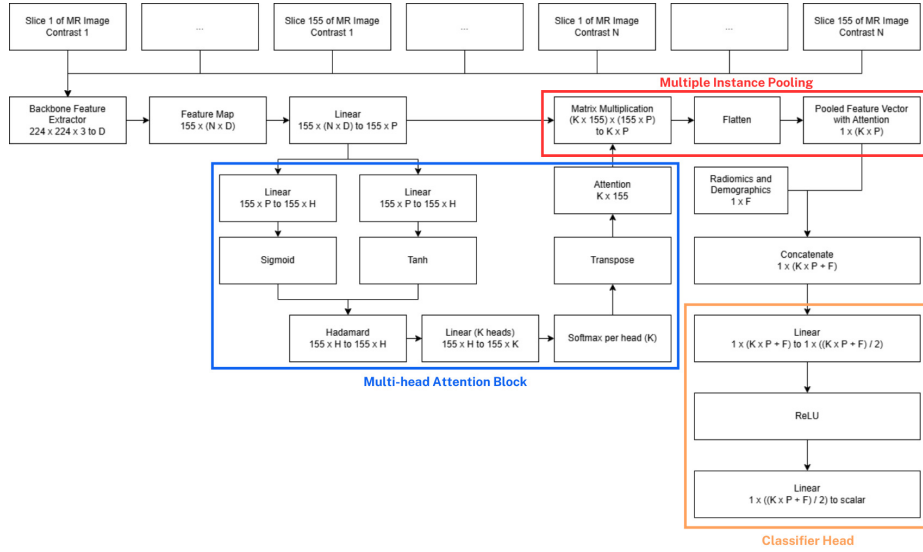


Fig. 1. Model Architecture

Several model configurations were considered in the study, which are listed in Table 1. For model configurations where only demographics and / or radiomics-based features were used, the XGBoost classifier from [6] and as implemented in [1] was instead used with the same loss function. Both the attention-based MIL neural network and the XGBoost classifier operate (i.e., take input and make inferences) at a patient level.

Table 1. Model configurations

Configuration	All MRI contrasts	T2w-FLAIR contrast	Demographics	Radiomics
1			✓	
2				✓
3			✓	✓
4	✓			
5	✓		✓	
6	✓			✓
7	✓		✓	✓
8		✓		
9		✓	✓	
10		✓		✓
11		✓	✓	✓

Note that there was class imbalance and as such, balanced class weights were used in all model training runs.

3.4 Hyperparameter Optimization

For hyperparameter optimization (HPO), Bayesian optimization (specifically, the Tree-structured Parzen Estimator or TPE, as described in [24]) was done using the Optuna package from [5]. The random seed was fixed and the batch size was also kept constant at 8. The following search spaces were defined for the HPO with TPE:

1. Attention-based MIL neural network as defined in Figure 1
 - (a) Projection size: 2^P where P is an integer between 6 and 10 if all MRI contrasts were used and an integer between 4 and 8 if only the T2w-FLAIR contrast was used
 - (b) Hidden embedding size for the attention block: 2^H where H is an integer between 5 and P if all MRI contrasts were used and integer between 3 and P if only the T2w-FLAIR contrast was used
 - (c) Dropout for both the first and last linear layers in the classifier head (shared parameter): A real number between 0.25 and 0.50

- (d) Temperature (a scalar by which each element in the attention matrices is divided): 1.0 to 5.0
 - (e) Learning rate for the optimizer: A real number between 10^{-5} and 10^{-3}
 - (f) Weight decay for the optimizer: A real number between 10^{-4} and 10^{-2}
2. XGBoost classifier
- (a) Max depth: An integer between 2 and 10
 - (b) Learning rate: A real number between 10^{-3} and 0.3
 - (c) Number of trees: An integer between 100 and 2000
 - (d) Subsample: A real number between 0.5 and 1.0
 - (e) Colsample: A real number between 0.5 and 1.0
 - (f) Lambda: A real number between 10^{-3} and 10
 - (g) Alpha: A real number between 10^{-3} and 10
 - (h) Minimum child weight: A real number between 10^{-2} and 10

Neural networks were trained for a maximum of 100 epochs, with early stopping implemented (i.e., a patience of 10 and minimum validation metric improvement of 0.002). To identify the best number of epochs (for re-training after HPO), the epoch at which the model achieved the best validation performance was recorded.

All ranges listed above include the endpoints of the range. The HPO for a neural network had 125 trials while XGBoost classifier had 500 trials each. Finally, the objective of the Bayesian optimization routine was to maximize the validation ROC-AUC.

3.5 Model Evaluation

Models were evaluated against the test set based on the following metrics: ROC-AUC, PR-AUC, accuracy, balanced accuracy, precision, recall, F1 score, and reversed Brier score, with the primary metric being ROC-AUC. Note that the reversed Brier score is simply 1 minus the Brier score. The confusion matrices were also constructed.

3.6 Software

Model training and evaluation were all done using Python 3 and additional Python packages. The `pytorch` package from [19] was used for training neural networks while the `xgboost` package was used for building XGBoost classifiers. As mentioned earlier, Optuna from [5] was used for HPO. For extracting radiomics-based features, the `mirp` package from [29] was used.

4 Results and Discussion

First, let us look at the HPO trial results for attention-based MIL neural network models, which are summarized in Figure 2. Notably, the largest validation ROC-AUC achieved was about 0.8823 while the smallest was about 0.6223. The

range of values is moderately large, suggesting that performance does vary across different models (or model configurations) and hyperparameters. However, all validation ROC-AUC values were above 0.5, suggesting the presence of a good signal from the features. The best validation performance was achieved by the All MRI contrasts-only model, although the All MRI contrasts + Demographics and All MRI contrasts + Demographics + Radiomics achieved nearly the same validation performance. On the other hand, T2w-FLAIR contrast-only models appear to perform the worst against the validation set, holding all other things constant.

Another interesting to note here is that demographics appear to be important when predicting IDH mutation status. Across various hyperparameter configurations and models, performance generally dipped when demographics features were excluded, while the impact of radiomics-based features were not as obvious. Overall, the models appear to be somewhat robust to hyperparameter configuration, but HPO is still clearly important for the model architecture described in Figure 1. Ultimately, the most promising model appears to be the one that uses all MRI contrasts, possibly augmented by demographics and radiomics-based features.

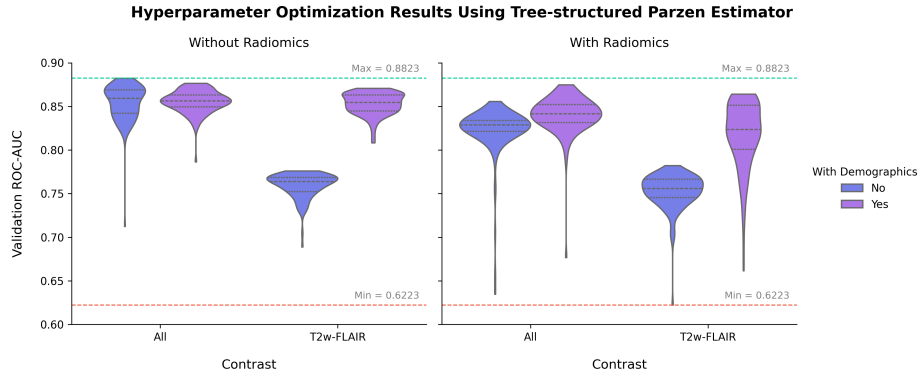


Fig. 2. HPO Trials for Neural Networks

After re-training the models on the combined training and validation sets with the best hyperparameters, their performance was evaluated against the held-out test set. The results are summarized in Figure 3.

Based on the test performance, the best model is clearly the one that uses all available features, as it has the best values across all metrics considered. That is, the one that uses all MRI contrasts, demographics features, and radiomics-based features achieved superior performance. While T2w-FLAIR contrasts can provide some value, it is clearly more beneficial to use all four MRI contrasts.

Notably, although it is not necessarily the best model, the model that uses only demographics features provide competitive performance. Despite using only two features, age and sex, it is able to achieve nearly the best performance in terms of almost all metrics. On the other hand, the worst performing model appears to be the one that uses radiomics-based features only.

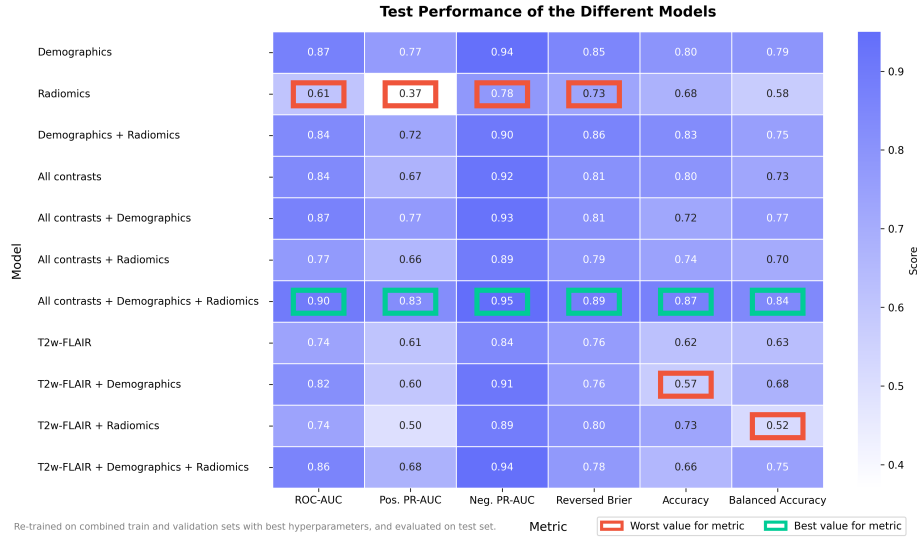


Fig. 3. Test Performance of Different Models

Overall, while MRI scans are still necessary, it appears that demographics features are extremely important when it comes to predicting IDH mutation status. Radiomics-based features, on their own, are not informative but its interaction with information from MRI scans and demographics can boost discriminatory performance. This suggests that using all available information is beneficial for predicting IDH mutation status but in the scarcity of resources, even looking at only the age and sex can already be informative. Together with these findings, looking at the beeswarm plot of the Shapley values across age and sex (Figure 4), we see that age appears to offer strong predictive power, such that younger patients are generally more likely to have IDH mutations. In general, females are more likely to have IDH mutations but the effect of sex is only marginal compared to age.

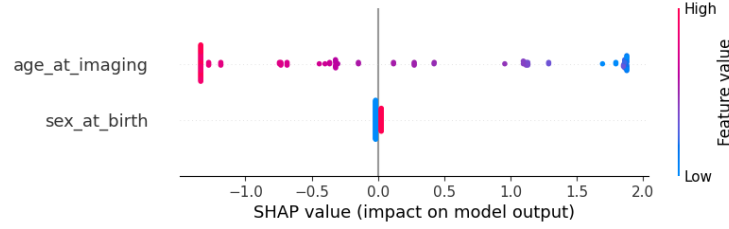


Fig. 4. Shapley Values from Demographics Features (demographics-only model)

Zooming in on the performance of the best model (all MRI contrasts + demographics + radiomics), we note that the precision, recall and F1 score for the positive class are 0.7692, 0.7692, and 0.7692, respectively. Meanwhile, for the negative class, we have 0.9104, 0.9104, and 0.9104, respectively. The overall accuracy and balanced accuracy are 87.10% and 83.98%, respectively. Moreover, the positive class and negative class PR-AUC are 0.8257 and 0.9547. Finally, the ROC-AUC and reversed Brier score are 0.8978 and 0.8884, respectively. The confusion matrix is also shown in Figure 5.

All in all, the performance of the model is excellent. The model exceeds random guessing baselines across all metrics, suggesting the presence of strong signals. Moreover, its ROC-AUC is about 0.90 (with a macro PR-AUC of about 0.89), which is excellent and is above the average pooled ROC-AUC of 0.8653 from different studies as found in [2]. From the confusion matrix, the diagonal counts dominate, with good performance for both classes. The Brier score (a proper scoring loss) is also comfortably below the baseline of 0.25, suggesting that the estimated probabilities themselves are sufficiently accurate (i.e., the classifier is sufficiently well-calibrated). Ultimately, these suggest that combining radiomics and demographics with attention-based MIL on MRI scans can predict IDH mutation status in glioma patients. Moreover, since the best model included MRI embeddings as features, the results also suggest that the pretrained model, BiomedCLIP, transferred reasonably well to the downstream task of IDH mutation status classification with radiomics and demographics.

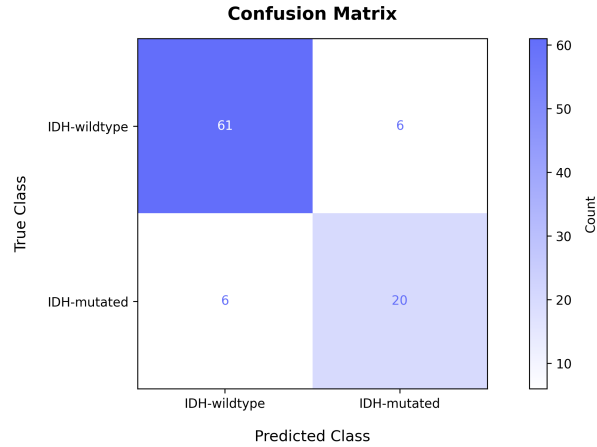


Fig. 5. Confusion Matrix for Best Model

5 Conclusion

In this paper, we constructed different models that considered different combinations of features from MRI contrasts, radiomics, and demographics. We were able to show that combining MRI contrasts with radiomics-based and demographics features has a strong potential for the non-invasive prediction of IDH mutation status, achieving an ROC-AUC of about 0.90 and an accuracy of about 87% (which are generally comparable to previous studies, exceeding the average). The combined-features model performed well, especially given the small dataset and frozen backbone. However, in the absence of available data, we also saw that age and sex alone (particularly, age) can already provide adequate performance for prediction (more so than radiomics-based features). Thus, it may be useful to explore other clinical data, specifically those that are “easy” to collect.

With that being said, there are several limitations and future potential lines of inquiry related to the study at hand:

1. The model architecture was limited. It may be beneficial to explore different architectures, still using attention-based MIL. For example, instead of performing the attention mechanism after concatenation, it may be better to individually create attention mechanisms for each contrast and only concatenating (the pooled vectors) after.
2. Although different combinations of feature groups were explored, granular feature selection was not performed here, which can potentially improve performance by reducing overfitting.
3. Radiomics-based features were not as useful in predicting IDH mutation status here, compared to other feature groups. However, that may be because only a pre-defined small set of radiomics-based features were considered.

4. Because of limited computational resources, cross-validation was not implemented here. Implement such can provide more reliable estimates of performance.
5. Demographics features may be worth exploring as predictors for IDH mutation status, as age and sex alone already provided competitive performance here.
6. Although it transferred reasonably well to the downstream task, the "backbone" here was completely frozen. Fine-tuning the backbone (or even training a backbone from scratch) can likely improve the performance.

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